

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
II Sem B.Tech (CBCS) Mid Semester Examinations - II, April 2023

CY1001-1 – ENGINEERING CHEMISTRY

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

	Unit – I	Marks	BT*	CO*	PO*
1.	a) What is a glass transition temperature? Discuss any four factors affecting the T _g .	5	L*2	2	1
	b) Explain the synthesis and applications of following polymers (i) Plexiglass; (ii) Epoxy resin	5	L2	2	1
2.	a) Explain the free radical mechanism of addition polymerization of vinyl chloride.	5	L2	2	1
	b) Thermosetting do not exhibit plasticity. Why?	2	L1	2	1
	c) Give the synthesis and applications of Carbon fibre.	3	L2	2	1
	Unit – II				
3.	a) Describe the experimental method to determine the hardness of water by complexometric method.	6	L2	2	1
	c) How is internal treatment for the softening of hard water carried out by (i) phosphate conditioning and (ii) colloidal conditioning?	4	L1	2	1
4.	a) Distinguish between sludges and scales?	2	L3	2	1
	b) Explain hot-lime soda process for the softening of hard water.	5	L2	2	1
	c) Explain the causes of boiler corrosion.	3	L2	2	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
